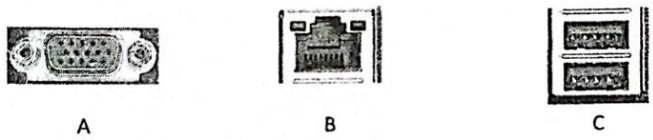
Higher National Diploma in Engineering (Electrical)

First Year, Second Semester Examination — 2022

EE 1212- Computer Technology

Question 01 (Total 20 Marks)

i. Identify the following ports of the motherboard and list a device that can be connected to each port. (06 Marks)



ii. Write down the technologies used in the following generations of computers. (04 Marks)

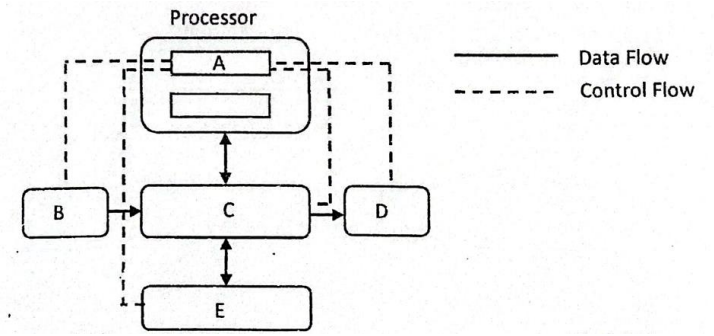
a. 1st generation

b. 2nd generation

c. 3rd generation

d. 4th generation

iii. The following figure shows the data flow and control flow of the computer system. Identify A, B, C, D and E devices/components. (05 Marks)



iv. List three (03) types of Microprocessors. (03 Marks)

v. Name any two (02) companies manufacturing microprocessors. (02 Marks)

Question 02 (Total 20 Marks)

i. What is an operating system? Give two (02) examples for operating systems. (04 Marks)

ii. Briefly describe them two (02) functions of operating systems. (04 Marks)

iii. Convert following binary numbers into octal numbers. (04 Marks)

a. 1001011101

b. 011000011

iv. Convert the following hexadecimal numbers into binary numbers. (04 Marks)

a. A34

b. 9F2

v. Calculate the following binary operations. (04 Marks)

a. -0011010 + 001100

b. 0011010001100

Question 03 (Total 20 Marks)

i. Draw a named block diagram of the microprocessor. (05 Marks)

ii. Briefly explain the following terms relevant to the microprocessors. (06 Marks)

a. Word length

b. Instruction set

c. Clock speed

iii. Name an operating system suitable for each device given below. (04 Marks)

a. PC

b. Laptop

c. Mobile Phone

d. Server

iv. List five (05) items/devices that you can find inside the system unit. (05 Marks)

Question 04 (Total 20 Marks)

i. Define the term modulation. (03 Marks)

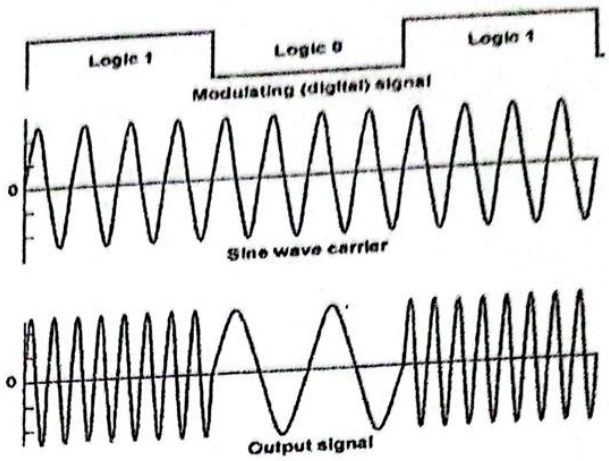
ii. Using a suitable diagram briefly explain amplitude modulation. (05 Marks)

iii. Fill in the blanks with the suitable terms (high/low). (06 Marks)

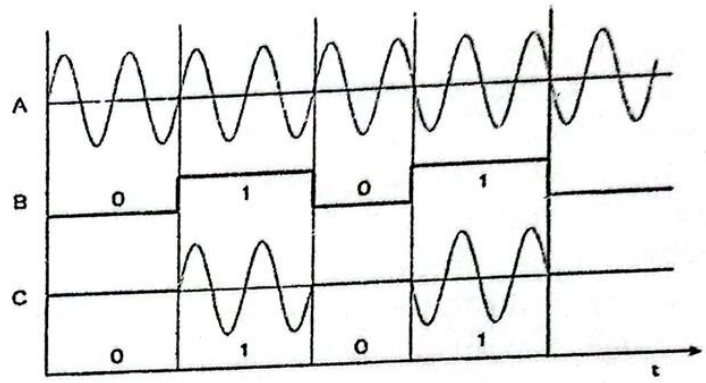
Feature	AM (Amplitude Modulation)	FM (Frequency Modulation)
Noise immunity	a.	b.
Bandwidth	c.	d.
Transmitted power	e.	f.

iv. Identify the modulation techniques used below. (06 Marks)

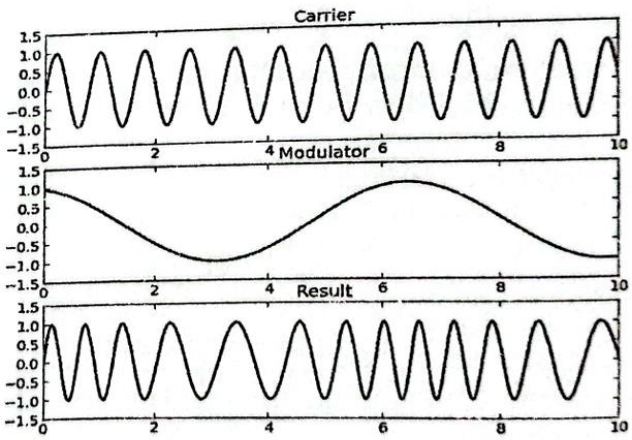
a.



b.



c.



Question 05 (Total 20 Marks)

i. Give two (02) advantages and two (02) disadvantages of computer networks. (04 Marks)

ii. List the most suitable data transmission media for the following situations. (03 Marks)

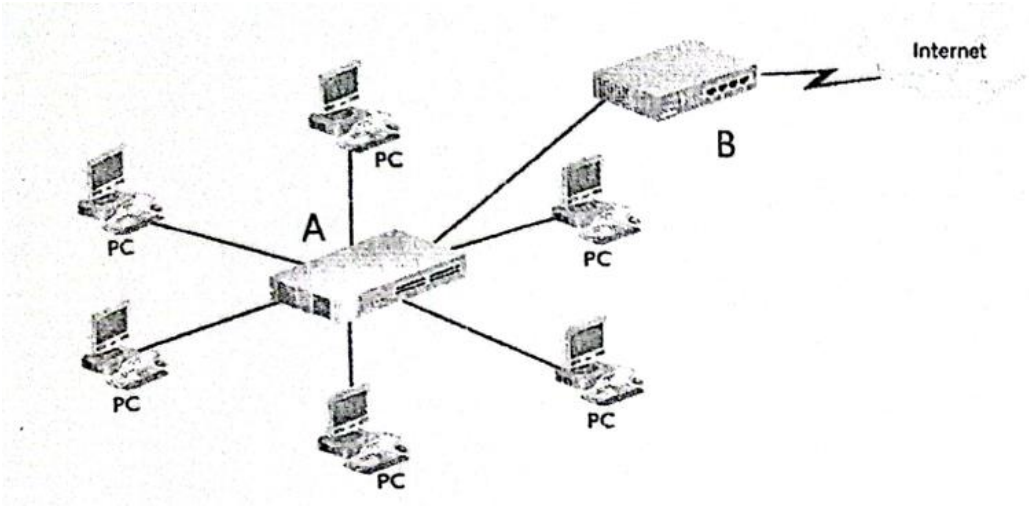
a. To set a WIFI zone

b. To set a network inside a computer lab

c. To interconnect two computer networks established in two cities

iii. What does it mean by the term protocol? Give two (02) examples for protocols. (03 Marks)

iv. Diagram of a computer network is given below. Identify the devices A and B. (02 Marks)



v. Briefly describe the following network devices. (04 Marks)

a. Ethernet Card/NIC

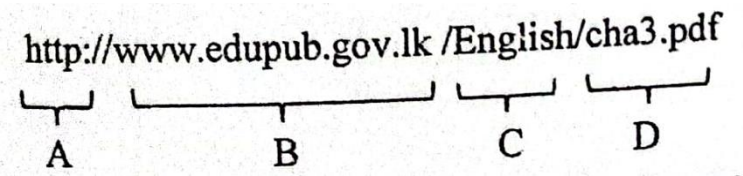
b. Router

vi. Personal Area Network (PAN) is a smaller type of network. Give one (01) device that can be connected to it and one (01) data transmission media that can be used for it. (04 Marks)

Question 06 (Total 20 Marks)

i. Relevant to a web page, briefly explain URL. (03 Marks)

ii. The structure of an URL is illustrated below, identify A, B, C and D. (04 Marks)



iii. Briefly describe the following addressing types of a computer. (06 Marks)

a. Physical address

b. IP address

iv. Mention whether the following IP addresses are valid or invalid. (04 Marks)

a. 10.50.01.0

b. 192.168.1.1

c. 200.400.2.34

d. 20:50:2:1

v. Identify the classes of following IP addresses. (03 Marks)

a. 192.168.1.1

b. 10.50.1.1

c. 160.10.1.1